Q10: Planning

- Due 16 Jan at 12:30
- Points 4
- Questions 4
- Available after 10 Dec 2025 at 17:00
- Time limit 60 Minutes
- Allowed attempts 5

This quiz is no longer available as the course has been concluded.

Attempt history

	Attempt	Time	Score
KEPT	Attempt 2	8 minutes	3 out of 4
LATEST	Attempt 2	8 minutes	3 out of 4
	Attempt 1	7 minutes	1 out of 4

Correct answers are hidden.

Score for this attempt: 3 out of 4
Submitted 10 Dec 2025 at 17:35
This attempt took 8 minutes.
Incorrect

Question 1
0 / 1 pts
Which of the following action schemas are NOT well-defined schemas in PDDL?

$$Action(Change(x, y, z)$$
$$PRECOND : Flat(x) \wedge Broken(z)$$
$$EFFECT : \neg Broken(z))$$

- [ ] because it contains a negation in its effect.

$$Action(Change(x, y, z)$$
$$PRECOND : (Flat(x) \vee Flat(y)) \wedge Broken(z)$$
$$EFFECT : Fixed(z))$$

- [x] because it contains a disjunction in its precondition.

$$Action(Change(x, y, z)$$
$$PRECOND : Flat(x) \wedge Broken(z)$$
$$EFFECT : FrontLeft(y) \wedge Fixed)$$

- [ ] because it contains more than one literal in its effect.

$Action(Change(x, y, z)$

$PRECOND: Flat(x) \wedge \neg Flat(y) \wedge Broken(z)$

$EFFECT: Fixed(z))$

because it contains a negation in the precondition.

Question 2
1 / 1 pts

Consider the following simplified Hanoi towers planning problem in PDDL.

$Init(On(B, One) \wedge On(S, Two) \wedge Clear(B) \wedge Clear(S) \wedge Clear(Three) \wedge Disk(B) \wedge Disk(S) \wedge CanPutOn(S$

$Goal(On(B, Two), On(S, B))$

$Action(Move(d, x, y)$
$PRECOND: On(d, x) \wedge Disk(d) \wedge Clear(d) \wedge Clear(y) \wedge CanPutOn(d, y)$
$EFFECT: On(d, y) \wedge \neg Clear(y) \wedge \neg On(d, x) \wedge Clear(x))$

Which of the following action sequences is a solution to the problem?

- $Move(S, Two, One), Move(B, One, Two) Move(S, Two, One), Move(B, One, Two)$
- $Move(S, Two, Three), Move(B, One, Two), Move(S, Three, B) Move(S, Two, Three), Move(B, One, Two), Move(S, Three, B)$
- $Move(S, Two, B) Move(S, Two, B)$
- $Move(S, Two, B), Move(B, One, Two) Move(S, Two, B), Move(B, One, Two)$

Question 3
1 / 1 pts

Consider the following planning problem in PDDL

$Init(At(P1, A), At(P2, B), At(P3, C), At(P4, D), At(P5, E), Pipe(P1, P3), Pipe(P3, P5), Pipe(P5, P2), Pipe(.$

$Goal(At(P1, D), At(P2, C), At(P3, A), At(P4, E), At(P5, B))$

$Action(Jump(x, from, to)$
$PRECOND: At(from, x) \wedge Pipe(from, to)$
$EFFECT: \neg At(from, x) \wedge At(to, x))$

Ignoring the precondition $Pipe(from, to)$ leads to defining a relaxed problem, whose solution can serve as a heuristic for solving the original problem via search. What holds for this heuristic?

- $h(n)$ is the number of misplaced letters.
- $h(n)$ is the number of misplaced letters divided by two.
- $h(n)$ is the number of misplaced letters multiplied by two.
- $h(n) = 0$

Question 4

1 / 1 pts

Consider the following planning problem in PDDL

$Init(At(P1, A), At(P2, B), At(P3, C), At(P4, D), At(P5, E), Pipe(P1, P3), Pipe(P3, P5), Pipe(P5, P2), Pipe(.$

$Goal(At(P1, D), At(P2, C), At(P3, A), At(P4, E), At(P5, B))$

$Action(Jump(x, from, to)$

$PRECOND : At(from, x) \land Pipe(from, to)$
$EFFECT : \neg At(from, x) \land At(to, x))$

At which level of the planning graph does the literal $At(P5, A)$ appear for the first time?

- Level S_2.
- Never.
- Level S_0.
- Level S_5.

Quiz score: 3 out of 4